

Latent World Models for Robot Planning: Attributing Failure to Representation, Search, or Cost

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Research Proposal · August 2026

Abstract

Robots that plan directly from camera images have converged on one architecture: an encoder maps images to a latent state, an action-conditioned predictor rolls that state forward, and an optimiser searches action sequences for the one whose predicted outcome minimises a distance to a goal image. Its four blocks are studied in separate literatures but evaluated by one joint statistic — end-to-end success rate — so failures cannot be traced to a cause. This proposal builds an instrument that makes the attribution tractable: a taxonomy of four failure surfaces, each carrying an explicit condition that would refute attributing a failure to it. Read against the literature, it exposes a methodological gap — nearly every published repair is evaluated under learned dynamics, so its gain cannot be separated from ordinary prediction error. A completed pilot runs the discriminating experiment for one surface and finds that removing prediction error does not close the gap, motivating a programme on costs and representations trained on the distribution the planner induces.

1 Introduction

Modern visual world models descend from the observation that a compact latent state suffices to train a controller inside a model’s own imagination (Ha and Schmidhuber, 2018) and can be planned in directly from pixels (Hafner et al., 2019). The field has since split along two axes that are rarely co-designed. One shapes the latent: reconstruction-based latent dynamics (Hafner et al., 2025), joint-embedding predictive architectures (Assran et al., 2025; LeCun, 2022), frozen pretrained visual features (Zhou et al., 2025), and value-aware latents (Hansen et al., 2024). The other searches over it: sampling-based MPC (Rubinstein, 1997; Williams et al., 2017), gradient-based planning, amortised control, and hierarchical decomposition (Zhang et al., 2026).

Failures are reported against the composition of the two, which is why the diagnostic literature disagrees about mechanism: prediction error and its exploitation (Bhamidipaty et al., 2026; Lambert et al., 2020), proposal coverage (Gao et al., 2026), action grounding (Chen, 2026; Seo et al., 2026), and cost geometry (Li et al., 2026; Wang et al., 2026) each claim to be the binding constraint, on different systems and under different conditions. Success rate cannot distinguish a bad encoder from a bad predictor, proposal distribution or cost, and prediction error is measured on a distribution the planner never visits. This proposal argues the disagreement is largely an artifact of protocol, and supplies the instrument and experiments that would settle it.

2 Failure surfaces and the methodological gap

We organise the problem around four *failure surfaces* (Table 1). The distinguishing feature is the third column: for each surface, the observation that would *refute* attributing a failure to it. A taxonomy without refutation conditions organises vocabulary; one with them organises experiments. One asymmetry is load-bearing: S1 is the only surface that can be closed *by construction*, by supplying exact dynamics to the candidates, which makes it the natural control condition for studying the other three.

Reading the literature through this taxonomy exposes the gap that is the analytical core of the proposal: *nearly every published repair for S2, S3 or S4 is evaluated under learned dynamics*, where S1

Table 1: Four failure surfaces, each with the condition that refutes attributing a failure to it.

Surface	Observable signature	What refutes an attribution to it
S1. Prediction error	A physically good action sequence is rolled out incorrectly	Failure persists when candidate dynamics are made exact
S2. Proposal coverage	No physically good candidate ever enters the population	Good candidates are demonstrably present and are not selected
S3. Transition realizability	Predicted transitions are inconsistent with the actions conditioning them	Failure persists when transitions are exact by construction
S4. Cost geometry / selection	Low-proxy-cost candidates have poor physical outcome; the gap widens with search	Proxy and physical rankings agree on the optimiser’s own realised populations

is open. A gain measured there cannot be attributed to the surface it claims to fix, because improving any component correlated with prediction quality helps whenever prediction is the binding constraint. The discriminating experiment — rerunning the repair with candidate dynamics held exact — has, to our knowledge, not been reported for any of them.

3 Preliminary results: a controlled pilot

A completed pilot shows the gap is consequential rather than pedantic. In tabletop Franka manipulation, candidate dynamics are made exact by snapshotting simulator state at each replanning step, executing every candidate and encoding the true rendered outcome, with encoder, optimiser, budget and horizon held fixed so that only the terminal cost varies. A privileged state-based cost then solves *push* in 64/64 episodes while the latent L_2 cost used by released systems solves 0/64, and a supervised state readout with 1.6–1.7 cm held-out decoding error — inside the 5 cm success radius — reaches at most 5/64 (Table 2).

Table 2: Terminal-cost comparison under *exact* candidate dynamics; 64 paired seeds, strict episode-end success, two released frozen encoders. The reference row is a privileged positive control for the search budget, not a deployable method.

Terminal cost	DINO push	DINO pick	JEPA push	JEPA pick
Simulator-state reference (control)	64/64	49/64	64/64	49/64
Latent L_2 (as deployed)	0/64	0/64	0/64	0/64
Supervised state readout	5/64	0/64	1/64	1/64

Conditioning on the optimiser explains why an apparently accurate readout fails. Readout and physical rankings agree moderately on initial candidate populations (Spearman 0.43–0.55) and collapse on the populations the optimiser itself produces after refitting (0.11–0.16), and the excess selection regret over a residual-matched permutation null exceeds the optimizer’s-curse baseline (Smith and Winkler, 2006) with all confidence intervals above zero. Eight interventions run in the same harness — off-policy-robust readouts, relearned adapters, LoRA fine-tuning to 94–100% grounding accuracy, ensemble-disagreement penalties, a grounded dynamics channel, amortised search-free control — each improved the fidelity quantity it targeted, and none moved contact planning outside the baseline band.

4 Research questions

RQ1. Which failure surface binds where? The diagnostic literature’s apparent disagreement may be an artifact of protocol — different systems, tasks and dynamics conditions — rather than a real disagreement about mechanism. Running the field’s existing S3 and S4 repairs under exact

candidate dynamics would settle, for each, whether its reported gain survives when prediction error is removed.

RQ2. Can a cost or representation be trained on the distribution the planner itself induces?

Every objective in the representation literature is fit on a distribution chosen by the experimenter; what a planner consumes is an *ordering* on a distribution chosen by the *optimiser*, after its search distribution has refitted. Whether an objective can be defined directly on that induced distribution is open, and the pilot’s recurring pattern — fidelity improves, planning does not — suggests it is the consequential question.

RQ3. Does the mechanism generalise beyond one optimiser and one representation family?

The pilot evidence comes from CEM and two frozen-encoder checkpoints. Because the mechanism depends on distribution refitting, it should transfer to MPPI and iCEM and behave differently under gradient-based and amortised control — a prediction, not yet a finding. The strongest competing explanation, that frozen features lack the spatial precision contact requires (Mur-Labadia et al., 2026), must be tested in the same harness, since the two imply opposite research programmes.

5 Significance, impact, and originality

Scientific significance. The proposal supplies what the field lacks: a way to say *why* a visual planner failed. Refutation conditions turn an informal vocabulary of failure into experiments with defined outcomes, making the literature’s competing diagnoses commensurable for the first time.

Corrective impact on an active literature. If the pilot’s finding generalises, a body of reported gains is misattributed: repairs credited to coverage, grounding or cost geometry may be repairing prediction error. Either outcome is informative — a repair that loses its benefit under exact dynamics was repairing S1; one that survives is a counterexample — and both redirect effort now spent on the wrong surface.

Practical impact on robot learning. The evidence points at a widely shared design assumption — that a representation faithful on a fixed dataset yields a cost usable by search — and shows it failing where manipulation is hardest: contact-rich tasks with centimetre-scale success criteria. Objectives defined on rankings within the populations a planner realises are the natural response and are untested at scale; and because the diagnosis targets the cost rather than the backbone, any resulting method composes with the frozen-encoder checkpoints the field already trains at great expense.

Methodological impact. Three quantities are cheap to compute and essentially never reported: candidate availability, selection regret against the best available candidate, and proxy–physical rank correlation at the first and last search iteration. Standardising them makes negative results interpretable, and is a contribution independent of any new method.

Originality. Three elements are, to our knowledge, new: refutation conditions attached to a failure taxonomy; the exact-dynamics control used as a general instrument for attribution rather than a one-off ablation; and measuring proxy–physical agreement *on the optimiser’s own realised populations* rather than on a fixed evaluation set — where the pilot’s agreement collapses and the mechanism becomes visible.

6 Methodology and work plan

The programme is a sequence of controlled comparisons in which one component varies and every other is pinned. Three commitments recur: (i) *exact candidate dynamics* as a control, closing S1 by construction; (ii) *matched harnesses*, holding encoder, optimiser, budget, horizon and success criterion

identical across arms; and (iii) *reporting conditioned on the optimiser*, measuring agreement and regret on realised populations at both the first and the final search iteration. Contact-rich and free-space cells are reported separately, because pooled suite-level rates conceal complete failure on the contact subset.

WP1 — Discriminating reruns of existing repairs (RQ1). Reimplement representative S3 and S4 repairs (inverse-consistency and action-grounding costs, reachability heads, learned terminal metrics) in the shared harness. For each: reproduce its reported gain under *learned* dynamics as a positive control, then rerun it unchanged under *exact* candidate dynamics. Outcome: a per-repair statement of how much of the gain survives removal of prediction error.

WP2 — Selection-aware objectives (RQ2). Develop costs and representations trained on the distribution the planner induces. Two families are compared against scalar regression onto true cost: group-relative *ranking* objectives over realised candidate populations, and a *decision-equivalence* criterion asking the latent to preserve the arg min over a family of goals — far weaker than reconstructing state, far stronger than predicting the next latent, and the reward-free analogue of TD-MPC2’s value supervision (Hansen et al., 2024). Training uses populations harvested from planner rollouts, refreshed as the induced distribution shifts.

WP3 — Generality and the competing explanation (RQ3). Two sweeps in the same harness: (i) representation families, including a dense-feature checkpoint (Mur-Labadia et al., 2026) that no published study has run through a cost-attribution protocol, adjudicating between the cost-geometry and representation-precision accounts; and (ii) optimisers — CEM, MPPI, iCEM and an amortised controller — testing whether the failure tracks distribution refitting. Hierarchical decomposition (Zhang et al., 2026) is a stretch arm.

WP4 — Evaluation protocol and release. Formalise the three reporting quantities and ship the harness, the exact-dynamics control and the diagnostic suite as an open benchmark with per-regime reporting; developed continuously alongside WP1–WP3.

Table 3: Indicative schedule for a two-year Master’s programme; WP4 runs continuously.

Period	Activity	Output
Months 1–6	Consolidate the pilot into the shared harness; add a second contact-rich task family; run the dense-feature checkpoint through the protocol (WP3-i)	Harness release; attribution paper
Months 7–12	WP1: reproduce three S3/S4 repairs under learned dynamics as positive controls, then rerun each under exact dynamics	Discriminating-experiment paper
Months 13–18	WP2: ranking and decision-equivalence objectives on planner-induced populations	Method paper; protocol spec
Months 19–24	WP3-ii optimiser sweep; offline real-robot check; benchmark release; thesis writing	Benchmark; Master’s thesis

Scope. WP1 and WP2 are the core and are sized to complete in full: the pilot harness, checkpoints and simulator infrastructure already exist, removing the setup cost that would otherwise dominate year one. WP3’s hierarchy arm can be dropped without loss to the thesis argument. Claims are stated at the scope the evidence supports: two checkpoint lineages and two task families, not latent planning in general.

Threats to validity. A null result under exact dynamics is uninterpretable without a reproduced positive control, so every WP1 arm carries one. Seed variance is real — one pilot arm produced 5/16 against a 0–2/16 band — so claims are made over paired environment seeds with clustered bootstrap intervals, and over checkpoint training seeds where feasible. Where an intervention is null,

the targeted quantity is reported alongside the planning outcome, so that “the repair did not work” stays distinguishable from “the repair did not engage”.

7 Resources and facilities required

The programme is computationally moderate: it trains no large backbone, and operates on released frozen checkpoints, small probes and terminal-cost heads.

- **Compute.** A GPU cluster with 40–80 GB-class accelerators (H100 or equivalent). The exact-dynamics condition dominates, since every candidate is executed and rendered — roughly 10^5 simulator rollouts per experimental cell. Indicatively 8–12 GPUs in bursts with SLURM-style batch scheduling; the pilot ran at this scale.
- **Simulation and storage.** MuJoCo-based manipulation suites (MetaWorld (Yu et al., 2020) and a second contact-rich suite) with privileged state access, required for the positive control and for regime stratification; 5–10 TB for cached latents, candidate populations and logs.
- **Datasets and checkpoints.** Publicly released world-model checkpoints across scales and lineages (Assran et al., 2025; Mur-Labadia et al., 2026; Terver et al., 2026; Zhou et al., 2025), plus an offline real-robot dataset for the transfer check. No new data collection is required.
- **Robot hardware (optional).** A Franka-class arm with wrist and third-person cameras would allow closed-loop confirmation, but is desirable rather than necessary: the exact-dynamics control is unavailable on real hardware, so the core experiments are inherently simulator-based and the arm serves only the final transfer check, for which an offline dataset substitutes.
- **Supervision.** Supervision spanning self-supervised representation learning and model-based control, since the central hypothesis lives at their interface. No fieldwork, human subjects or restricted data are involved.

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